



Project Report

SECJ3623: Mobile Application Programming
Semester I, Session 2025/2026

Project Title: Smart Recipe Finder
Group Name: NULL

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1. Project Background & Problem Statement

1.1 Background

With the increasing use of smartphones, many daily activities such as shopping, communication, and information sharing are now performed through mobile applications. Cooking is one of the daily activities where people frequently search for information online, especially food recipes. However, recipes are often scattered across websites, social media, notebooks, or messaging applications, making them difficult to organize and retrieve efficiently.

Small food business owners and cooking enthusiasts also face challenges in storing their recipes digitally while sharing them with others. A centralized mobile platform is needed to allow users to browse, save, and share recipes conveniently using their mobile devices.

1.2 Problem Statement

Currently, there is no simple mobile platform that allows users to browse recipes without login, while also supporting recipe sharing and management in one system. Many existing platforms require registration before access, which discourages casual users. Additionally, recipes stored manually or across different platforms may be lost, duplicated, or difficult to manage.

Therefore, a mobile application is required to provide an organized, user-friendly, and accessible recipe management system that supports different user roles and basic recipe-sharing functionality.

2. Objectives & Project Scope

2.1 Objectives

The objectives of this project are:

- To develop a mobile application that allows users to browse and search food recipes easily.
- To provide a platform for users to share and store recipes digitally.
- To enable the system owner to manage users and recipe content effectively.
- To support guest access for viewing recipes without requiring login.

2.2 Project Scope

The scope of this project includes:

- Development of a mobile application using **Flutter**.
- Backend services using **Firebase** for authentication and data storage.
- Support for three user roles: **Admin (Owner), User/Creator, and Guest**.
- Implementation of basic CRUD (Create, Read, Update, Delete) functions for recipes and users.

The project does **not include** payment processing, food delivery services, or advanced analytics features.

3. Design Solution

3.1 Design Solution

Smart Recipe Finder Application System is designed to meet the functional and usability requirements of all user groups through a structured and scalable architecture. The system includes a dedicated administrative module that enables administrators to manage recipe content and user accounts efficiently. This module supports content creation, modification, approval, and deletion, as well as user monitoring and moderation, ensuring data integrity, security, and content quality across the platform.

For end users, the application provides optimized browsing and search functionality through indexed recipe data and advanced filtering mechanisms based on ingredients, cuisine type, preparation time, and dietary constraints. Users can store selected recipes in a personalized favourites repository, which is linked to their user profile for persistent access. This design improves system performance and enhances user experience by minimizing navigation time and search complexity.

The system also supports user-generated content by allowing registered users to submit their own recipes via a structured input interface. Submitted content is stored in the central database and made available to other users, subject to validation or administrative approval if required. This functionality promotes community interaction and continuous content growth within the application.

To accommodate non-registered users, the system implements a guest access layer that allows read-only access to recipe data without authentication. This approach reduces entry barriers while maintaining system security by restricting write operations to authenticated users only.

Finally, the application adopts a mobile-first, responsive user interface design that ensures consistency, accessibility, and ease of use across mobile devices. The interface leverages intuitive navigation, adaptive layouts, and touch-optimized controls to support efficient interaction for all users, regardless of technical proficiency.

4. Stakeholder and User Identifications

4.1 Stakeholder Identification

Stakeholder is the group of individuals who have an interest in developing and the successful of the Smart Recipe Finder Mobile Application.

Stakeholder of this project is development team, project owner and end users. The development responsible on the designing, implementing, integration and testing the application. The project owner will ensure the applications met the user requirements need and expectation. End user is the user who interacts with the application to find, create and view the recipes.

4.2 User Identification

The three (3) targeted users for the mobile platform application are as follows:

1. Admin (Owner)

Each application required admin/owner to maintain and managing the users and recipes which is applied on this project also.

2. User / Creator

The users who enjoy browse, search, bookmark, and share recipes.

3. Guest

The users who only want to view recipes without logging in.

The users need for this application:

1. Admin needs a system to manage recipe content and user accounts.
2. Users need a simple and fast way to browse, search, and save recipes.
3. Users want to share their own recipes with others.
4. Guests need the ability to view recipes without mandatory registration.
5. All users require a simple and intuitive mobile interface.

5. UML Diagram

5.1 UML Diagram (Actor Only)

The UML diagram, actor only illustrated the interaction between user and Smart Recipe Finder mobile application. The main actor is user or creator who perform action such as registering, logging in, viewing recipes details, searching for recipes, creating recipe details, saving favourite recipes via bookmarks, and edit profiles. An admin actor manages the application contents by managed user, and managed recipes by adding, editing, and deleting recipes and users. The guest actor only allows to search and view recipes without mandatory registration or logging in.

This diagram helps visualize system functionality and user interaction at a high level.

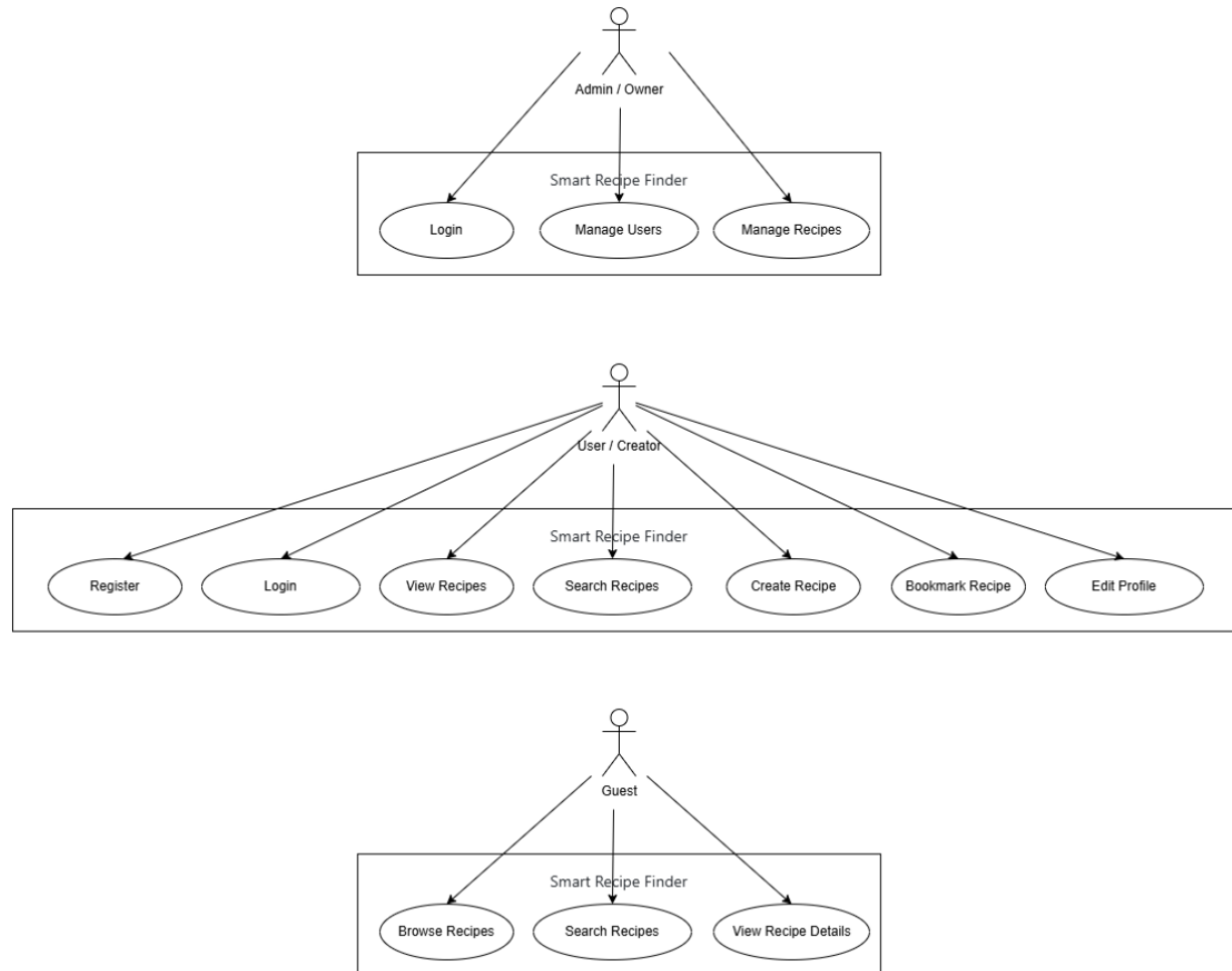


Figure 5.1: UML diagram (actor only) for Smart Recipe Finder Application System.

6. System Flow

6.1 System Flow Description

System flow is role-based flows that control how user interact with Smart Recipe Finder Application based on designation role such as admin, user/creator, and guest. The overall flow is supported by the Firebase services for the authentication, data storage, and media handling.

6.1.1 Admin Flow

1. The admin logs in through Firebase Authentication.
2. The system verifies the admin role and grants access to administrative functionalities.
3. Admins can:
 - View, activate, deactivate, or remove user accounts.
 - Add, edit, update, or delete recipes to ensure content accuracy and quality.
 - Monitor overall application content.
4. Admin actions directly modify user and recipe records stored in Cloud Firestore.
5. The admin oversees system maintenance and ensures proper functioning of the application.

6.1.2 User/Creator Flow

1. The user registers or logs in using Firebase Authentication.
2. After successful authentication, the system verifies the user's role and grants appropriate access.
3. Users can:
 - Browse and search recipes stored in Cloud Firestore.
 - Bookmark favorite recipes, which are saved to the user's profile data.
 - Add and share personal recipes by submitting:
 - Recipe text details,
 - Recipe images (uploaded to Firebase Storage)
4. Uploaded images are stored in Firebase Storage, and their URLs are linked to the corresponding recipe records in Firestore.
5. Users can manage and update their personal profile information.
6. All user actions are validated by role-based access control to prevent unauthorized operations.

6.1.3 Guest Flow

1. A guest accesses the application without authentication.
2. The system grants read-only access to public recipes stored in Cloud Firestore.
3. Guests can browse and view recipe details.
4. When a guest attempts to bookmark a recipe or share their own recipe, the system prompts them to register or log in.
5. Upon successful registration or login, the guest role transitions to a User/Creator role.

6.2 System Flow Diagram

Flow diagram illustrates as system role based.

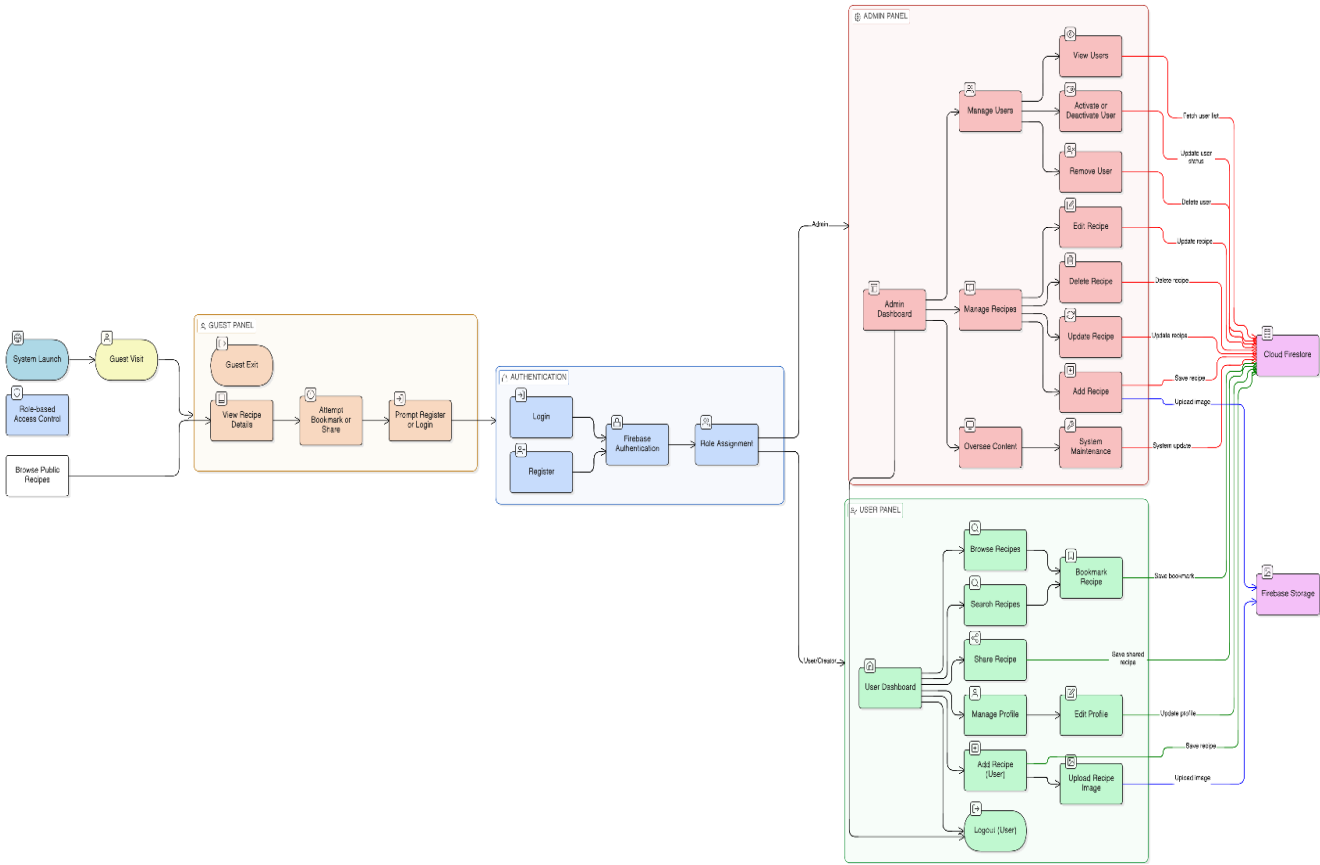


Figure 6.1: Flow diagram for Smart Recipe Finder Application System.

7. Low Fidelity and High Fidelity

7.1 Low Fidelity

7.1.1 Admin Low Fidelity

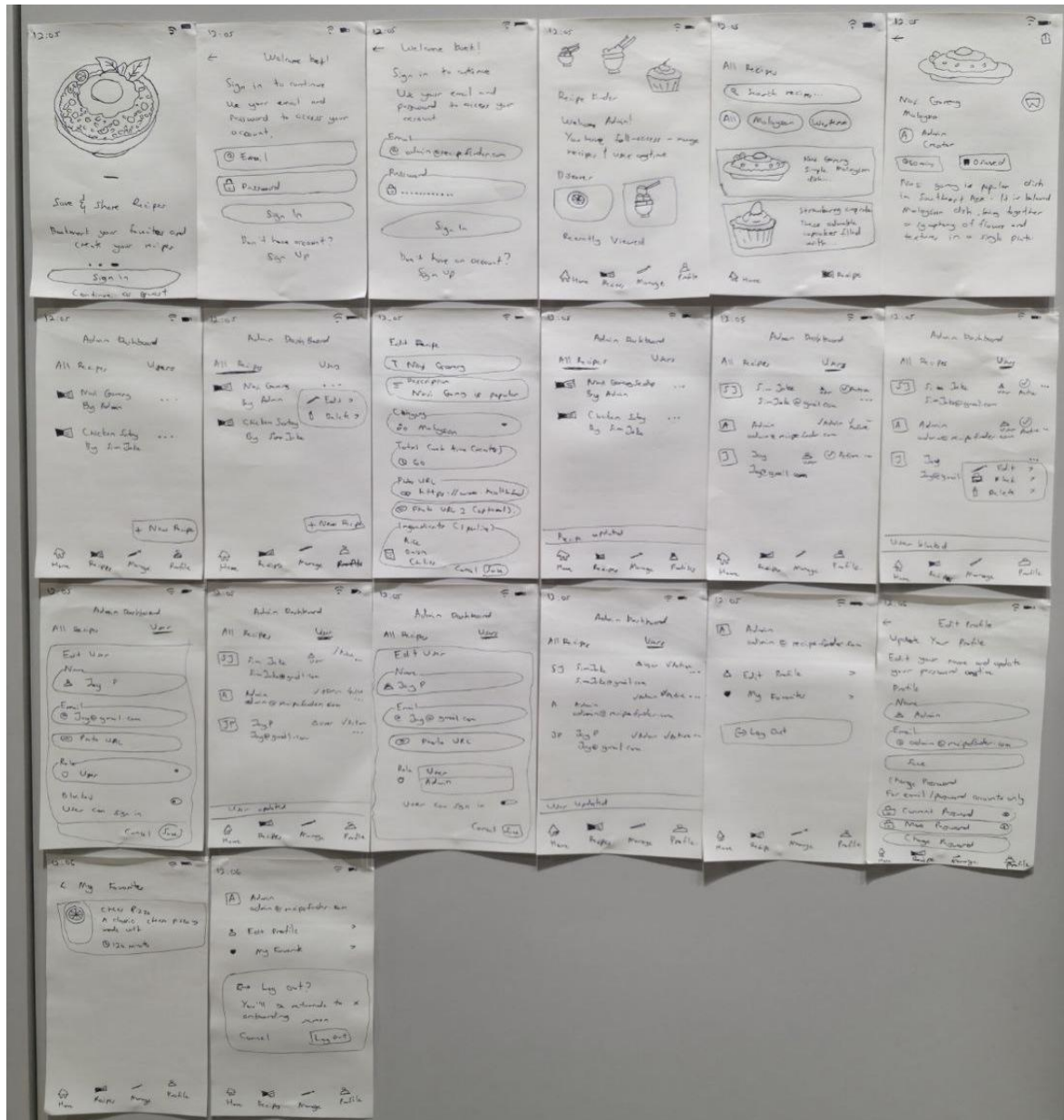


Figure 7.1: Admin low fidelity for Smart Recipe Finder Application System.

7.1.2 User/Creator Low Fidelity

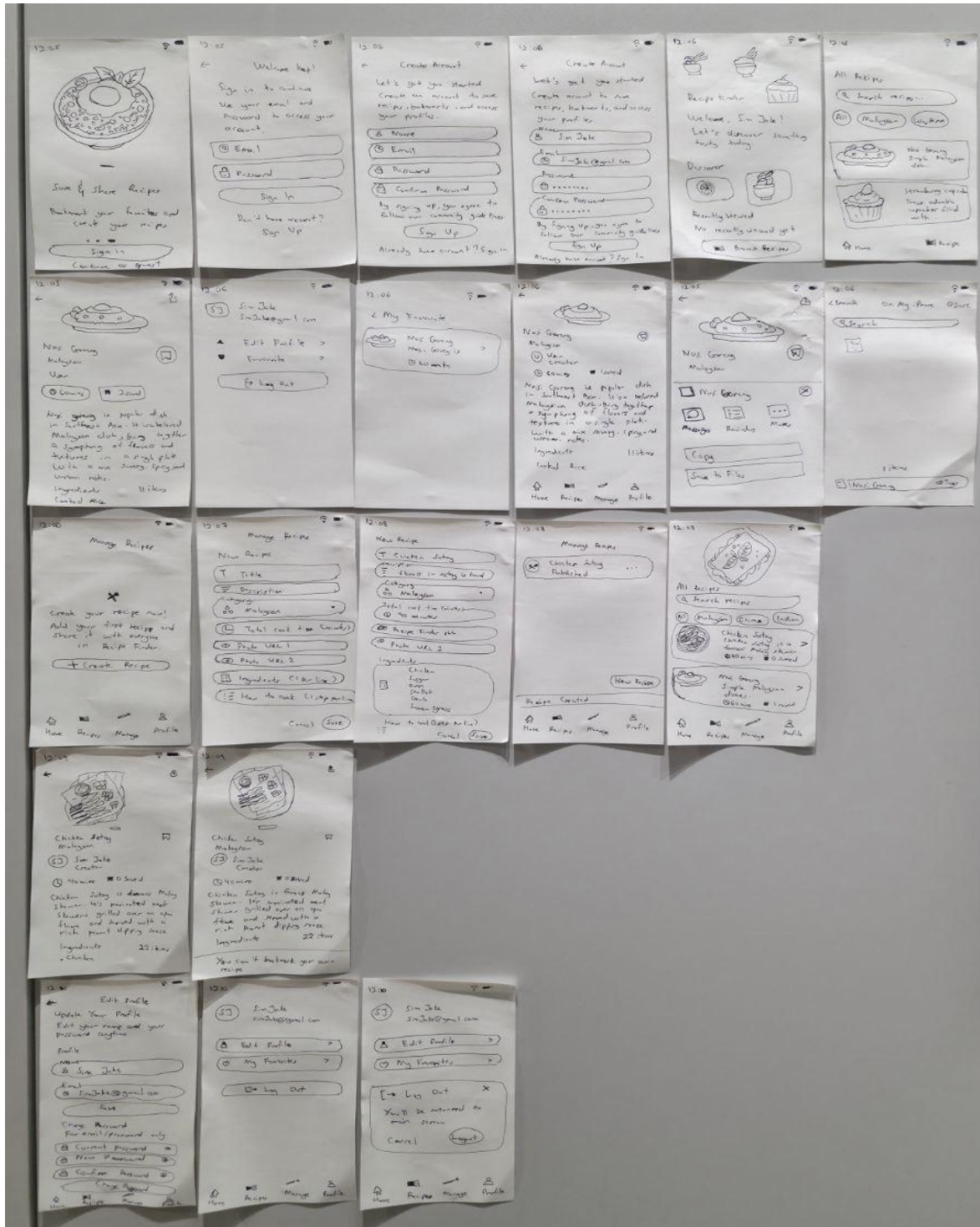


Figure 7.2: User/Creator low fidelity for Smart Recipe Finder Application System.

7.1.3 Guest Low Fidelity

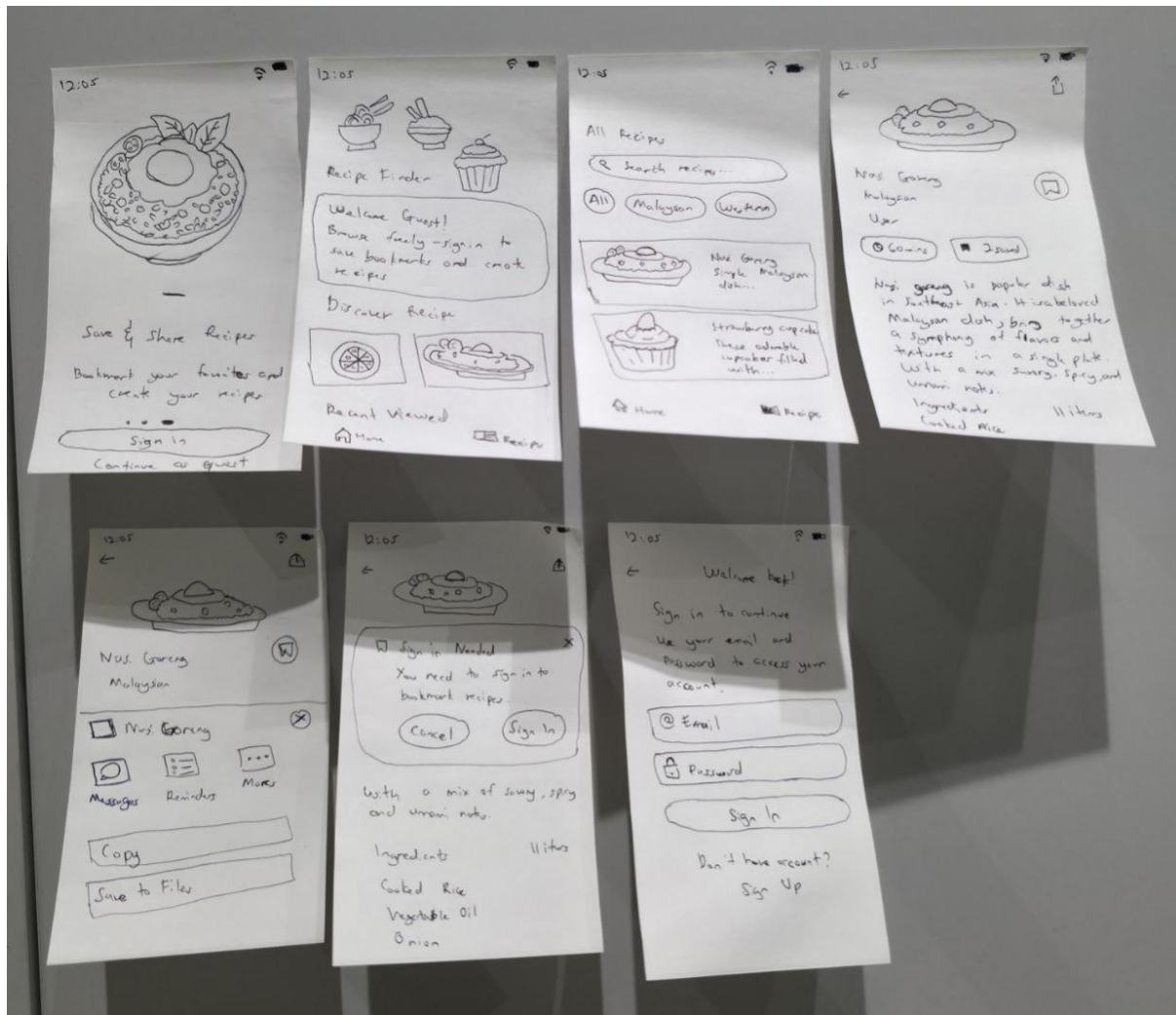


Figure 7.3: Guest low fidelity for Smart Recipe Finder Application System.

7.3 High Fidelity

7.3.1 Admin High Fidelity

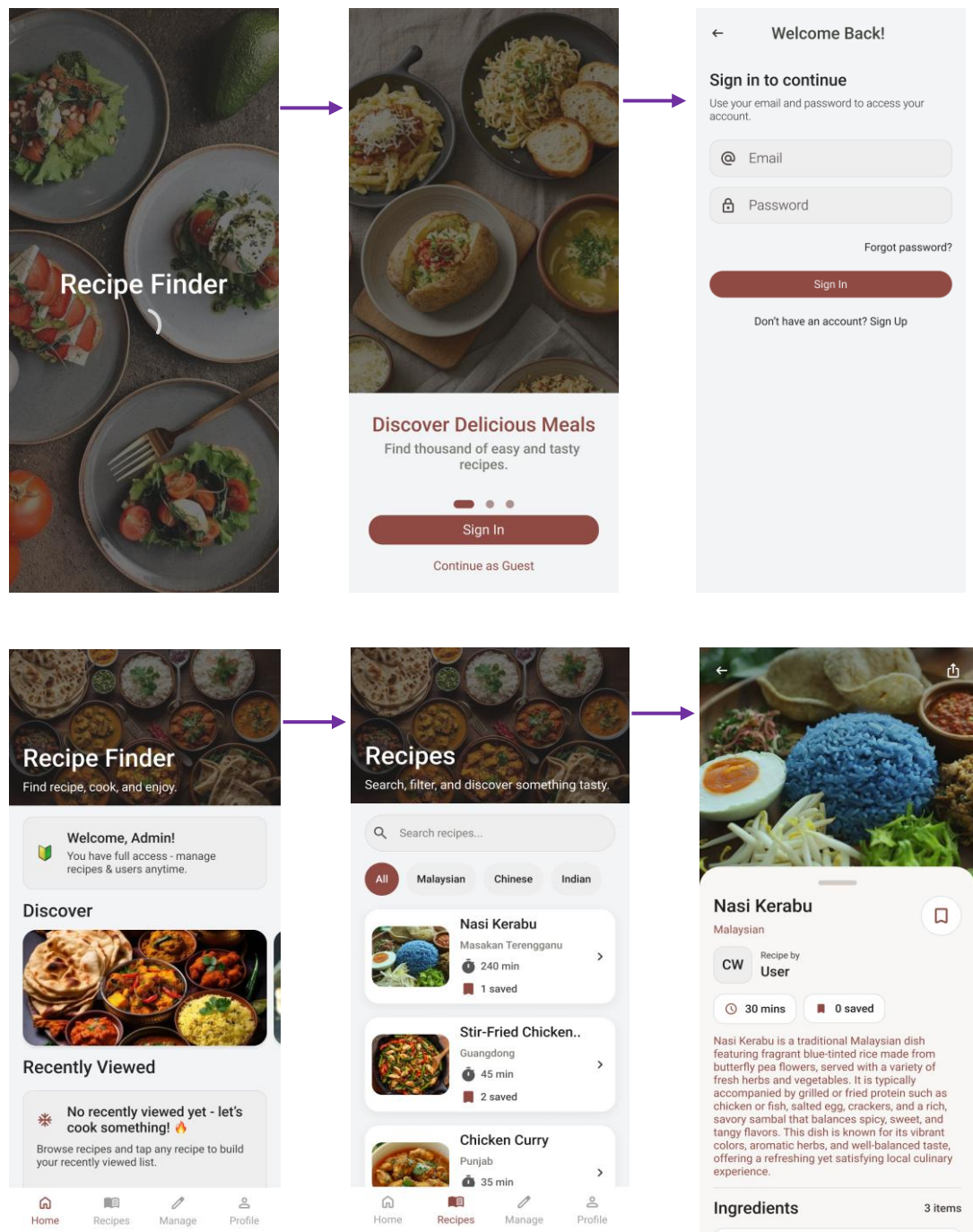


Figure 7.4: Admin high fidelity for Smart Recipe Finder Application System.

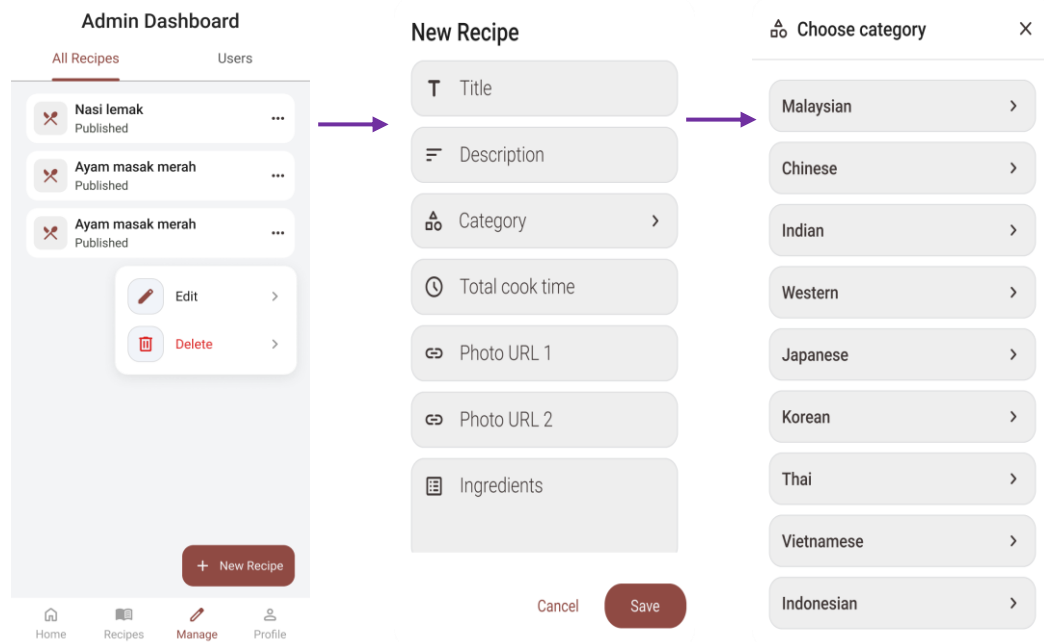


Figure 7.5: Admin high fidelity for Smart Recipe Finder Application System.

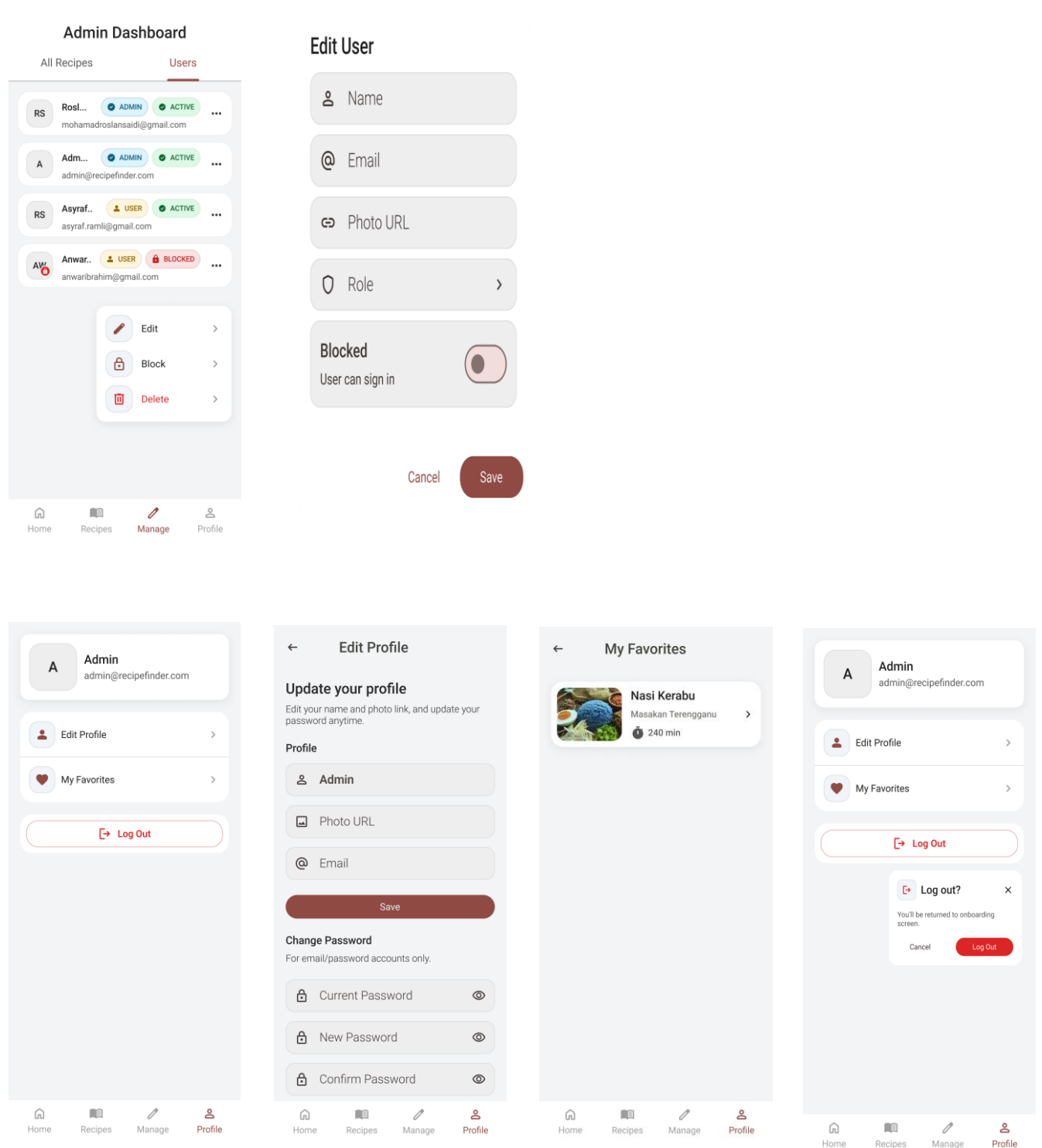


Figure 7.6: Admin high fidelity for Smart Recipe Finder Application System.

7.2.2 User/Creator High Fidelity

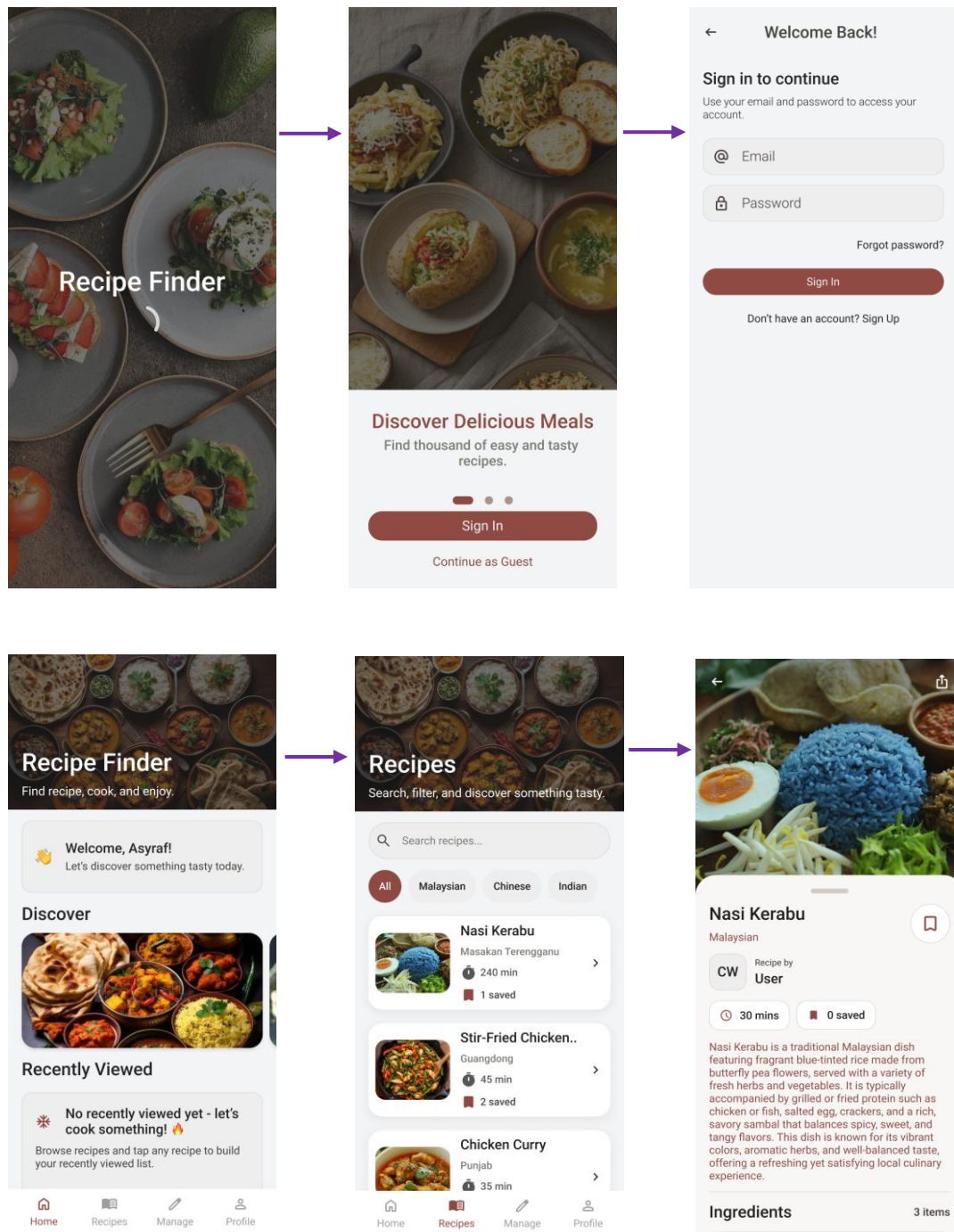


Figure 7.7: User/creator high fidelity for Smart Recipe Finder Application System.

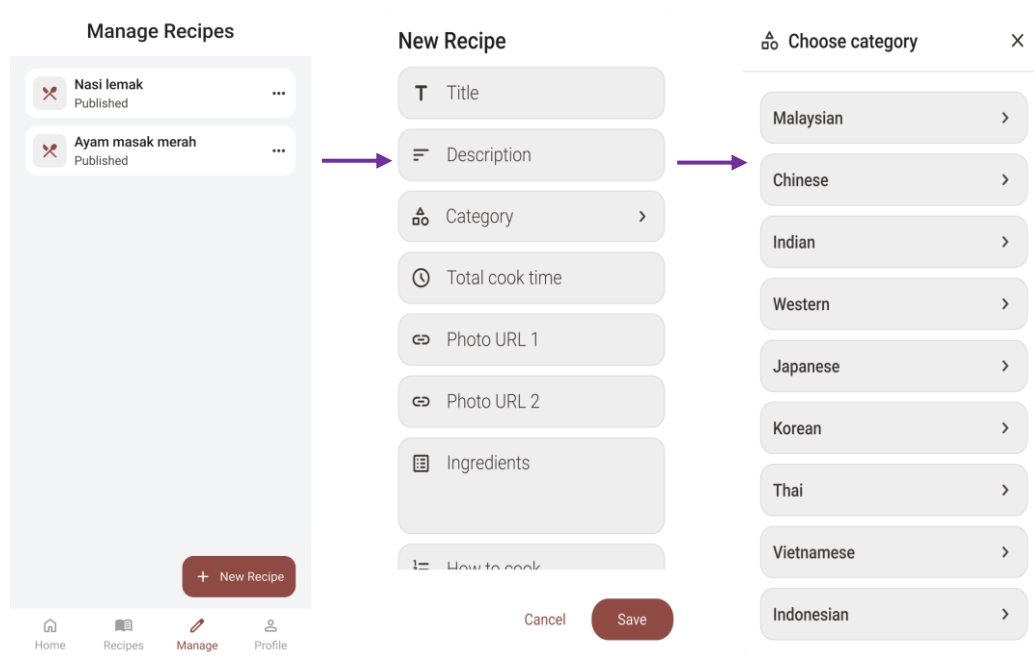


Figure 7.8: User/creator high fidelity for Smart Recipe Finder Application System.

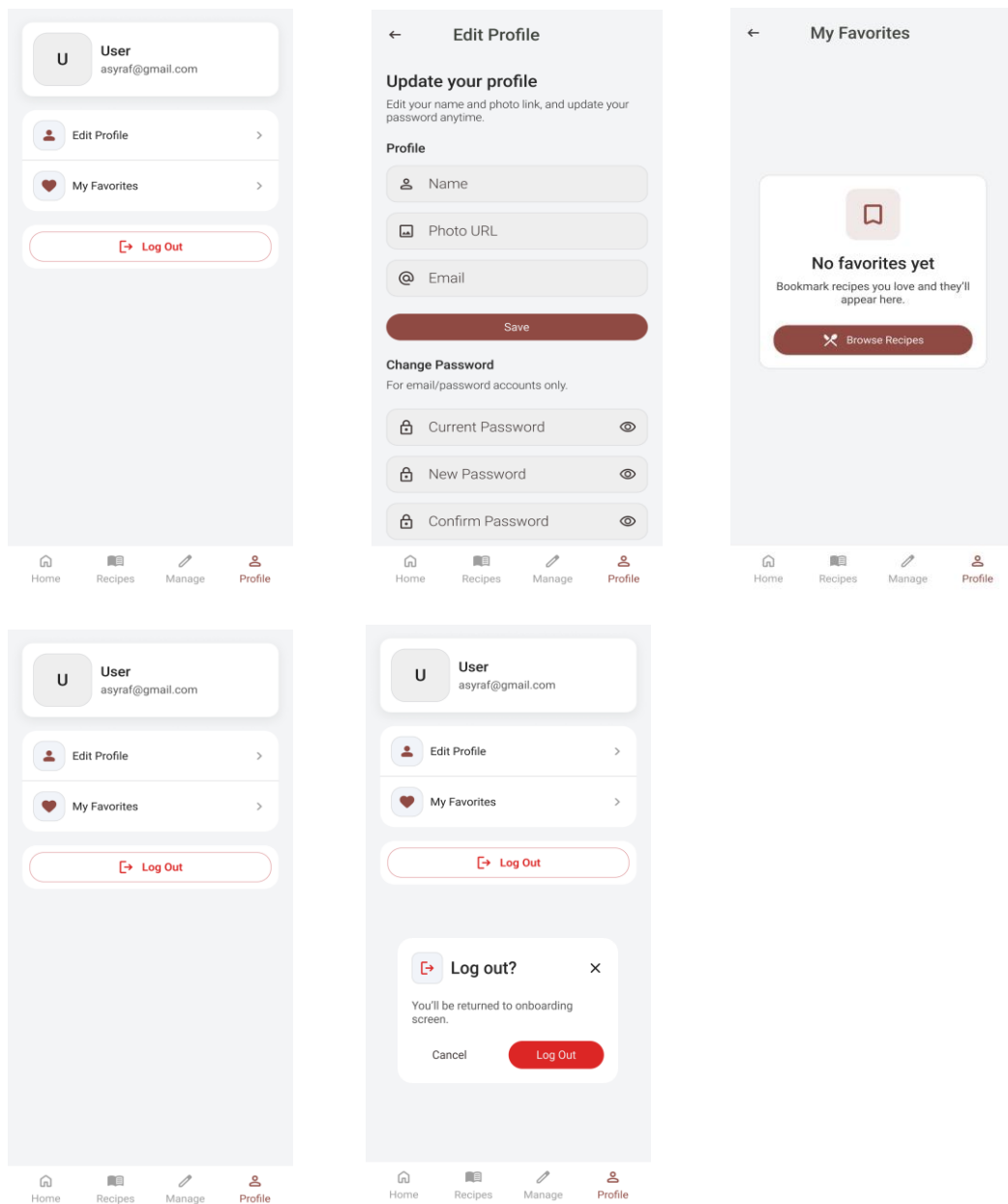


Figure 7.9: User/creator high fidelity for Smart Recipe Finder Application System.

7.2.3 Guest High Fidelity

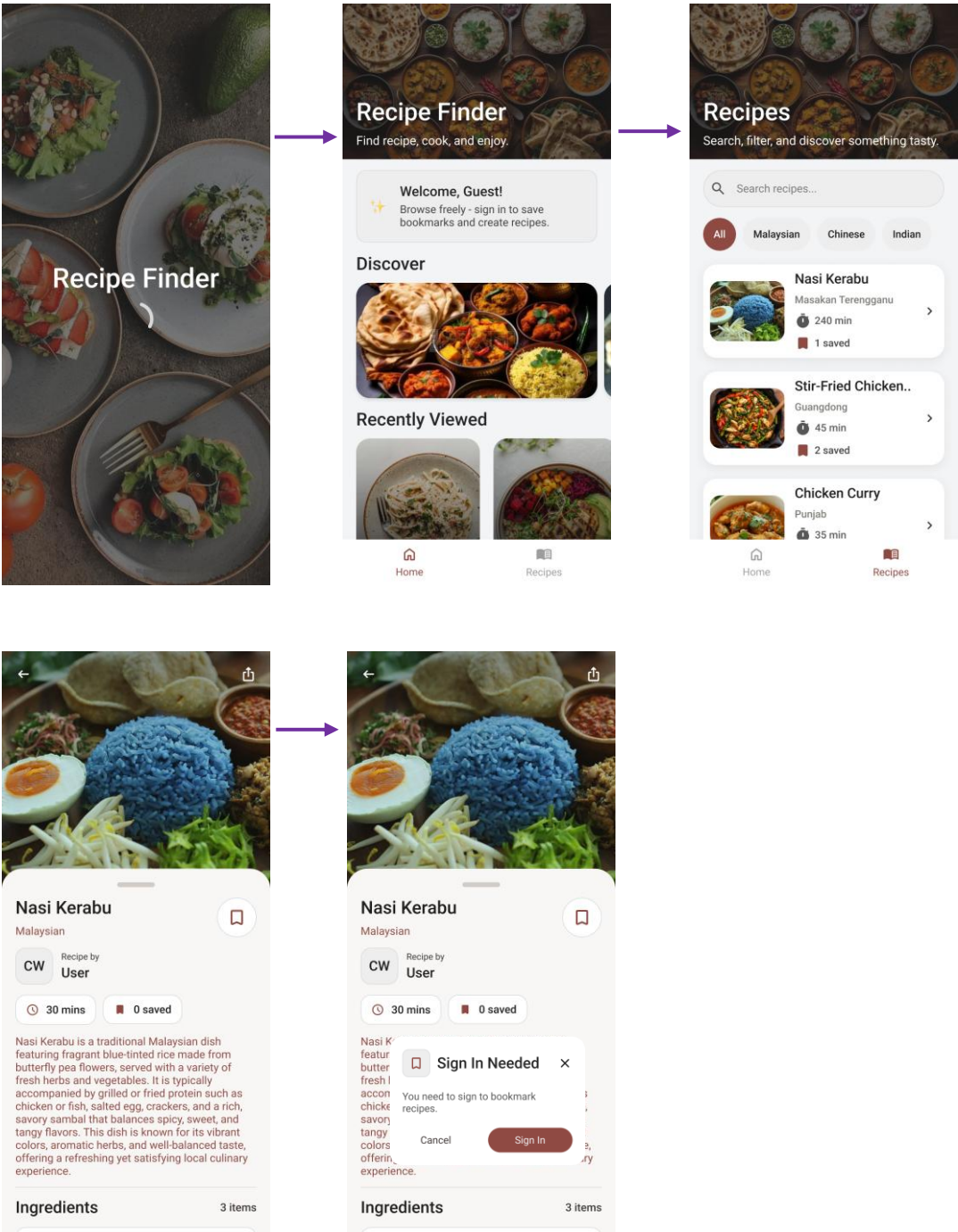


Figure 7.10: Guest high fidelity for Smart Recipe Finder Application System.